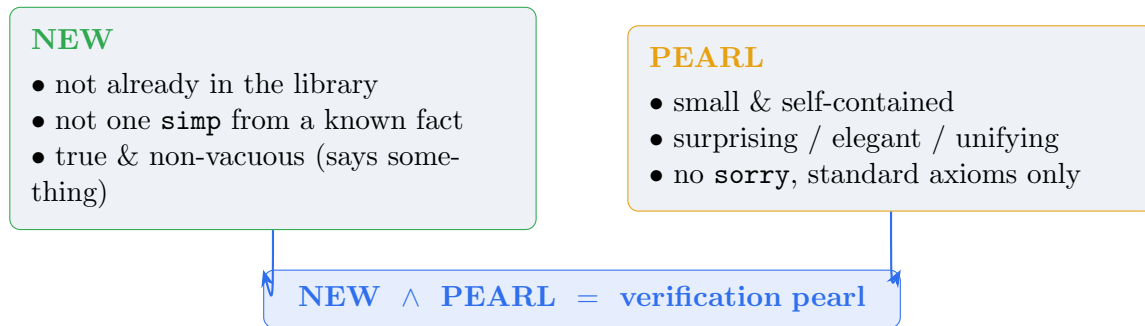


The Anatomy of a Verification Pearl

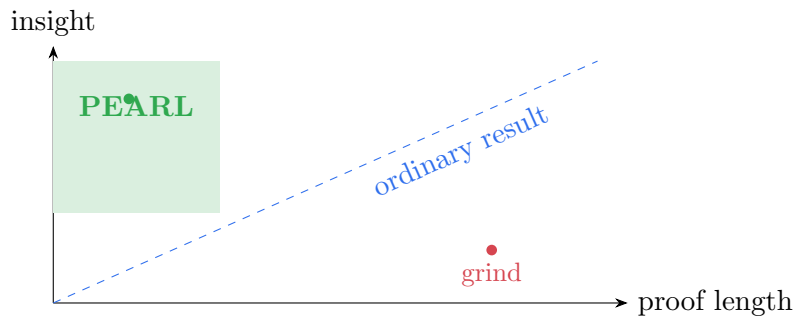
A tiny statement, a short proof, a big “aha”, fully machine-checked.

1. What a pearl must be

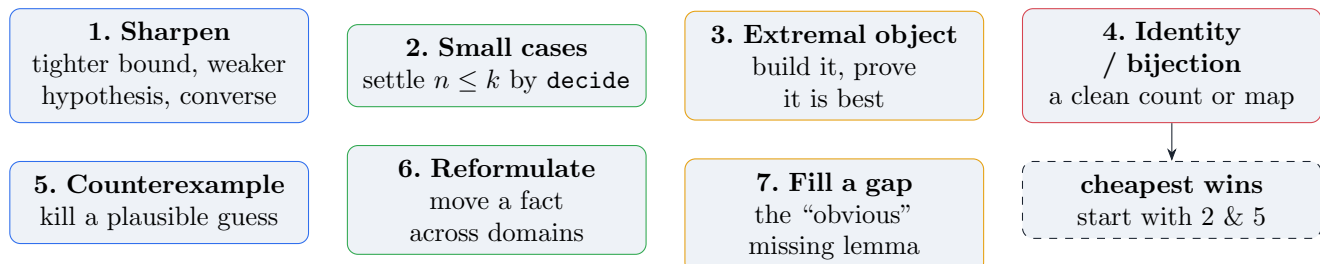
Two boxes must both be ticked. *New* on the left, *beautiful* on the right.



The signature of a pearl: the “aha” is much bigger than the proof.



2. Seven roads to a pearl



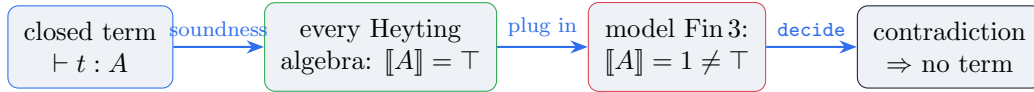
3. The worked pearl: Peirce's law has no term

Claim. In the simply typed λ -calculus there is *no* closed term of type

$$((p \rightarrow q) \rightarrow p) \rightarrow p \quad (\text{Peirce's law}).$$

It is a *classical* tautology, yet *intuitionistically* unprovable — so no program has that type. This is road **5** (counterexample) proved by road **6** (reformulate into algebra).

The plan, on one line:



No normalization theory is needed: one induction plus one finite check.

4. The engine: syntax maps into semantics

Curry–Howard reads *types as propositions* and *terms as proofs*. The interpretation $\llbracket - \rrbracket$ is a structure-preserving bridge: an arrow type becomes Heyting implication $\Rightarrow$, a context becomes a meet $\sqcap$.

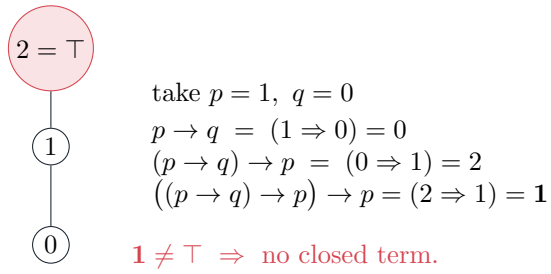
$$\begin{array}{ccc}
 \text{Syntax} & \xrightarrow{\llbracket - \rrbracket} & \text{Heyting algebra } H \\
 \\
 A \rightarrow B & \xrightarrow{\llbracket - \rrbracket} & \llbracket A \rrbracket \Rightarrow \llbracket B \rrbracket \\
 \vdash t : & & \uparrow \leq \llbracket A \rrbracket \\
 \Gamma & \xrightarrow{\llbracket - \rrbracket} & \bigwedge_{C \in \Gamma} \llbracket C \rrbracket
 \end{array}$$

Soundness is exactly the statement that the square commutes: a derivation $\Gamma \vdash t : A$ forces $\llbracket \Gamma \rrbracket \leq \llbracket A \rrbracket$ in *every* H . For a closed term $\Gamma = \emptyset$, so $\top \leq \llbracket A \rrbracket$, i.e. $\llbracket A \rrbracket = \top$.

soundness rule of thumb
 provable $\Rightarrow$ true in *all* models
 so: false in *one* model $\Rightarrow$ not provable

5. The refuting model Fin 3

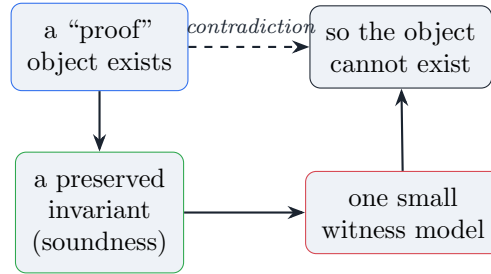
The smallest Heyting algebra that is not Boolean has three values $0 < 1 < 2$ (with $2 = \top$). Peirce's law evaluates *below* $\top$, which is impossible for a closed term.



Contrast: the identity $\lambda x. x : A \rightarrow A$ *does* exist, so the type system is not empty — the pearl is a real distinction, not vacuous.

6. The reusable pattern

Every counterexample-by-model pearl has the same shape. Swap the boxes for your setting.



Small statement $\rightarrow$ one clean invariant $\rightarrow$ one finite witness $\rightarrow$ done.